1. I2S0/TDM PROGRAM-AUDIO LVDS LINK

CMS_AUDIO_OFFSHEET_PORT
U900

SN65LVDS047PW
U901

SN65LVDS047PW
U902

PCA9060
U903

J901 routes with 4x50 Ω RIB (12). Route the 100 ohm differential pairs with the assigned interleaved returns.

SN65LVDT20 includes the 40Ω series resistor termination; do not add a duplicate for-next resistor.

2. DEDICATED LOW-NOISE HEADSET RAILS

LP5907MFX-3.3
U904

LP5907MFX-1.8
U905

HEADSET_PWR[0..2] wire are dedicated to the codec/headphone path. HEADSET_PWR[0..2] comes from the GPIO expander.

Codec phase and baseline decoupling follow Radxa CM5 IO V2.2 validate Control protection datasheet values before release.

3. ES8316 I2S1 HEADSET CODEC

AUDIO_BUS_TDM_CONTROL
J901

ES8316
U912

ES8316 is controlled on HS_I2C_SCL/HS_I2C_SDA. No GPIO expander interrupt is required; status is polled.

HEADSET_PWR[0..2] and HEADSET_AMP_EN default low through 100K pulldowns so the jack remains muted during boot.

Sequence enable codec rails, configure ES8316, enable TPA6132A2, then ramp software volume.

4. TPA6132A2 HEADPHONE AMPLIFIER

TPA6132A2RTE
U913

10nF and 10nF snap the amplifier for 0 dB gain. ON remains software-controlled for pop-free mute.

5. CTIA HEADSET JACK, MIC BIAS, DETECT, AND ESD

PJ-327A_CTIA
J910

TPD4S012DQDRQ4
U914

CTIA mapping is tested. The PJ-327A supplier footprint still requires mechanical verification before PCB release.

6. HEADSET I2C AND CONTROL POLICY

PJ-327A is wired CTIA: tip left, ring 1 right, ring 2 ground, sleeve microphone
TPA6132A2 provides the requested dedicated stereo headphone amplification
I2S1 drives an ES8316 headset codec based on the Radxa CM5 IO V2.2 reference
I2S0/TDM program audio is buffered over LVDS to the separate AUDIO-8X8 board

ProComm

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